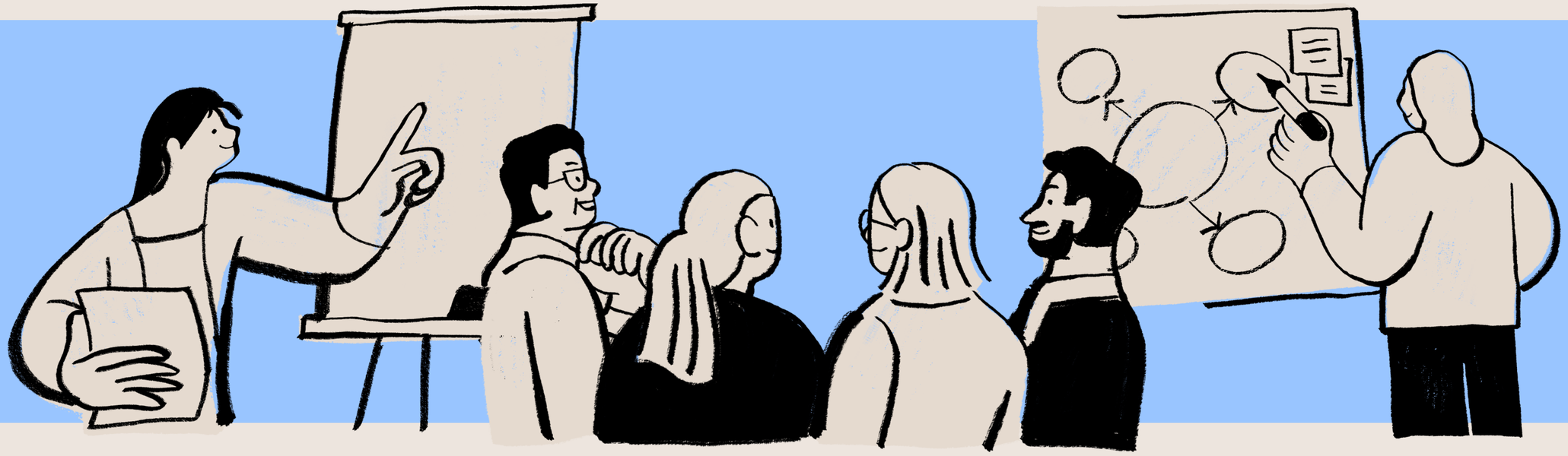
Group 1: Pooja (0013), Shreya (0089), Xana (0322)

BINARY LOGISTIC REGRESSION

Predicting College Student Placements



INTRODUCTION

This study uses **Binary Logistic Regression** to analyse which academic, skill-based, and extracurricular factors significantly influence whether a student gets placed.

This model is used when the **outcome (dependent variable) is dichotomous** like yes/no or success/failure, and in our case, placed/unplaced.

We don’t use Linear Regression because:

1) LRM Predicted values aren’t bounded between 0 and 1, and probabilities cannot be, say, 1.3 or -2.

2) Non-constant variance (heteroskedasticity), since linear regression assumes constant variance, but the variance of y depends on the mean (Var(y)=p(1−p)

3) Nonlinear relationship between predictors and probability, because the relationship between x and P(y=1) is typically sigmoid.

DATA OVERVIEW

College_ID	IQ	Prev_Sem_	CGPA	Academic_Performance	Internship_Experience	Extra_Curricular_Score	Communication_Skills	Projects_Completed	Placement
CLG0030	107	6.61	6.28	8	No	8	8	4	No
CLG0061	97	5.52	5.37	8	No	7	8	0	No
CLG0036	109	5.36	5.83	9	No	3	1	1	No
CLG0055	122	5.47	5.75	6	Yes	1	6	1	No
CLG0004	96	7.91	7.69	7	No	8	10	2	No
CLG0015	96	5.26	5.32	7	No	5	8	0	No
CLG0071	123	6.68	6.58	5	No	7	8	2	Yes
CLG0096	111	8.77	8.76	7	No	3	1	2	Yes
CLG0097	92	6.47	6.33	9	No	7	8	5	No
CLG0057	108	8.82	8.6	4	No	5	9	1	No
CLG0063	93	8.73	8.9	2	Yes	5	6	0	No
CLG0077	93	6.23	6.51	8	No	5	7	4	No

Data Size

10 Columns
& 10,000 Rows

Link

Access the dataset here!

Source

Kaggle: College Student Placement Factors Dataset

Statistical Technique:

- Binary Logistic Regression

Dependent Variable:

- Placement (0 = Not Placed, 1 = Placed)

Independent Variable:

- IQ (normally distributed around 100)
- CGPA (range: ~5.0 to 10.0)
- Prev_Sem_Result (range: 5.0 to 10.0)
- Academic_Performance (scale: 1 to 10)
- Internship_Experience (0 = No, 1 = Yes)
- Extra_Curricular_Score (score from 0 to 10)
- Communication_Skills (scale: 1 to 10)
- Projects_Completed (count between 0 to 5)

Tool used:

- PSPP



METHODOLOGY



STATISTIC	VALUE	INTERPRETATION
Log Likelihood	4372.52	Measures unexplained variation
Cox & Snell R^2	0.37	Model explains about 37% of variation
Nagelkerke R^2	0.62	Indicates a strong model fit: explains 62% of variation in placement outcome



MODEL SUMMARY

CLASSIFICATION TABLE

Classification Table

Observed			Predicted		
			Placement		Percentage Correct
			0	1	
Step 1	Placement	0	8011	330	96.0%
		1	651	1008	60.8%
	Overall Percentage				90.2%



The model accurately predicts overall placement status 90.2% of the time. It performs extremely well in predicting non-placement (96%), and fairly well for placement (61%). This slight imbalance indicates there are more non-placed students in the dataset

PREDICTORS

VARIABLE	B	p-value	Exp (B)	Interpretation
IQ	0.11	0.000	1.11	Significant, every 1-point increase in IQ increases odds of placement by 11%
Prev_Sem_Result	0.21	0.131	1.23	Not significant (p > 0.05); slight positive trend but weak effect
CGPA	1.05	0.000	2.86	Highly significant, each unit rise in CGPA increases odds of placement ~3 times
Academic_Performance	-0.01	0.621	0.99	Not significant, negligible effect
Internship_Experience	0.04	0.628	0.628	Not significant,slight positive but weak
Extra_Curricular_Score	-0.02	0.190	0.98	Not significant, small negative trend
Communication_Skills	0.65	0.000	1.91	Highly significant; each improvement in communication skill nearly doubles placement odds
Projects_Completed	0.68	0.000	1.97	Highly significant; each additional project skill nearly doubles placement odds
Constant	-29.19	0.000	2.10E-013	When all predictors are 0, placement likelihood is extremely low (virtually zero)

LOGIT MODEL

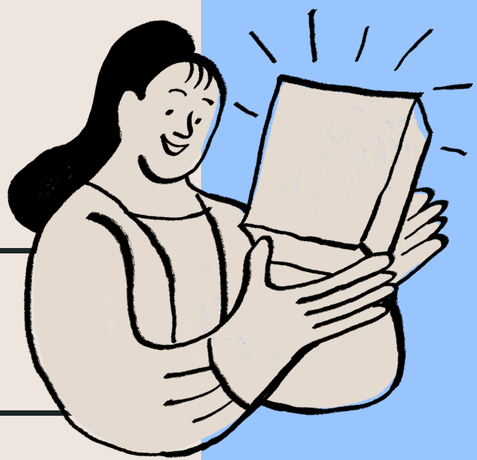
$$\text{logit}(p) = -29.19 + 0.11IQ + 0.21Prev_Sem_Result + 1.05CGPA - 0.01Academic_Performance + 0.04Internship_Experience - 0.02Extra_Curicular_Score + 0.65Communication_Skills + 0.68Projects_Completed$$

Probability Form

$$p = \frac{1}{1 + e^{-\text{logit}(p)}}$$



EQUATIONS



Low performing student

IQ = 90
Prev_Sem_Result = 50
CGPA = 5.0
Academic_Performance = 50
Internship_Experience = 0
Extra_Curricular_Score = 20
Communication_Skills = 1
Projects_Completed = 0

→ Linear predictor = −3.790 (approx)

Hence, **Predicted probability p≈0.0221** (≈2.2% chance of getting placed)

Average performing student

IQ = 110
Prev_Sem_Result = 65
CGPA = 6.5
Academic_Performance = 70
Internship_Experience = 1
Extra_Curricular_Score = 50
Communication_Skills = 3
Projects_Completed = 1

→ Linear predictor = 4.355 (approx)

Hence, **Predicted probability p≈0.9873** (≈98.7% chance of getting placed)

High performing student

IQ = 130
Prev_Sem_Result = 80
CGPA = 8.5
Academic_Performance = 90
Internship_Experience = 2
Extra_Curricular_Score = 80
Communication_Skills = 5
Projects_Completed = 3

→ Linear predictor = 13.705 (approx)

Hence, **Predicted probability p≳0.9999** (practically 100% chance of getting placed)

EXAMPLE PREDICTIONS



CONCLUSION

The logit model shows that CGPA, communication skills, projects completed, and IQ significantly increase placement chances. The model fits well, explaining 62% of the variation in placement outcomes with an overall accuracy of 90.2%, indicating strong predictive power and reliable performance.



THANK YOU!